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Design and Implementation of Wireless IoT Device for Women's Safety

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IoT and Embedded Systems

Design and Implementation of Wireless IoT Device for Women's Safety

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ABSTRACT

Abstract—Nowadays, women face imminent risk of sexual harassment, rape and molestation in the streets as male predators roam around in search of their potential targets. In this precarious scenario, women needs to be protected in the street for their safety. To address these challenges, in this paper, we propose a wearable device that will protect and enhance the safety of women. This goal is achieved by analyzing the user's pulse rate and oxygen measurements and updating them on the Blynk IoT web server. The user's pulse-rate and blood oxygen saturation is determined by acquiring raw heart-rate and oxygen values from a pulse oximeter and oxygen sensor installed at the heart of the device. Real-time monitoring of user data is achieved by transferring sensor data to the Blynk IoT web server. This wireless device is specifically programmed to continuously monitor the user's pulse-rate and oxygen values and react in case of a danger situation. It does so by detecting the abnormal pulse-rate and oxygen values, after which it sends the alert text message along with the location and live video streaming to the guardian's mobile device via the Tasker app. Meanwhile, the alarm starts ringing. Besides, we validate our system architecture using different real time test cases. Moreover, we present a comparison of our proposed architecture with existing literature's.

CCS CONCEPTS

• **Human Computer Interaction**;

KEYWORDS

Live Streaming; Women Safety; ESP-32 Camera; GPS; Pulse Oximeter GPS; Pulse Oximeter; Heart- Rate Sensor

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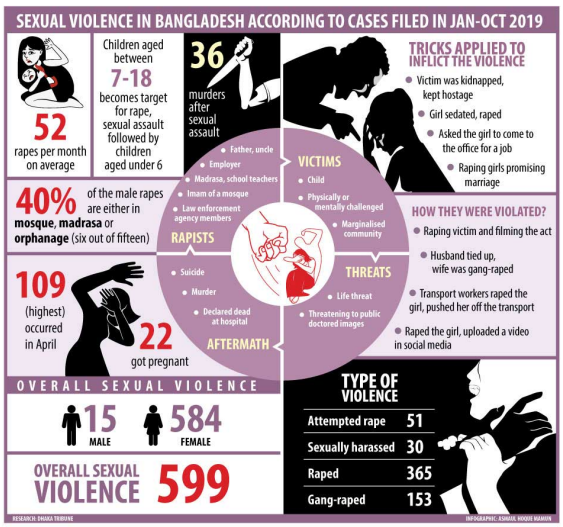

Figure 1: Sexual Violence Statistics in Bangladesh [21]

1 INTRODUCTION

The Daily Star published a news on Nov 7, 2021 about a brave girl from Bangladesh, who jumped off a moving bus to avoid being raped [22]. This brave girl was travelling in a bus and could sense immi- nent danger. When the driver attempted to rape her she jumped of the bus and was saved by another car driver. Just like this scenario, women face immense threat of sexual harassment in the streets, as rapists target them to become their victim. So, their safety is a global issue at the moment. According to a report by Odhikar, a Bangladeshi human rights organization, gang rape cases in the street have nearly tripled from under 100 in 2007 to nearly 300 by the year 2021. In the year 2021 alone, around 3700 women and children in Bangladesh were victims of sexual abuse including rape and murder [21] (as shown in Figure 1).

In today's society, such cases are all too common, which is unacceptable. We need more courageous girls like her who can protect themselves from these criminals. To empower women, we researched in this area and decided to build a safety device which can be used in these kinds of scenarios. On the other hand, even if the situation is against her, the device can take action using her vital signs such as heart-beat and oxygen levels. For this, we need a wireless device. That is when IoT comes for solving the problem, and that is what our research focuses on.

As of today, many IoT devices exist in the marketplace for the safety of women, employing various techniques. Choosing the most appropriate technique to build a cost- effective reliable safety device is one of the biggest challenges for researchers and manufacturers. Here, time delays in sending alert text messages to the parent's mobile device are directly related to the effectiveness and reliability of the device. Furthermore, for successful implementation of the wireless safety device, proper use of IoT, cloud infrastructure, and other applications is necessary. As such, several existing research studies focus on IoT and other architectures based on sending alert text messages to the support system after detecting an assault. The study in [6] proposes a product that, when pressed, broadcasts the victim's whereabouts to the family, friends, and the police. Other studies [7–9] propose multimodal cost-effective security paradigms for women who may be subjected to offensive risks using deep sensing technologies. Unfortunately, these studies propose and analyze only real-time test cases based on certain scenarios and lack applicability and effectiveness in a real assault.

We propose a wireless IoT device that will provide safety for women while they are in any dangerous situation. The devices here communicate using IoT. The working model here, is a device that sends alerts via text using the GSM cellular method, relay's location coordinates via GPS, and generates an alarm. Wirelessly transferring sensor data to an open-source Cloud Platform which allows data monitoring of current data [8]. This equipment is set up to continuously monitor women's vital parameters like pulse rate and temperature [3], and activate automatically if a dangerous scenario arises. It is easy to wear as the material is flexible and non-irritating for the skin and adjustable enough to ensure a comfy fit. This new technology is built in such a way that it serves its purpose to the fullest, which is ensuring women's safety and providing an active lethal form of self-defense. Based on our study, we make the following set of specific contributions in this paper:

- We explore several existing women safety systems and find out drawbacks of the system.
- We propose a women safety system comprising of unique features in our system including a live video streaming option and a reset button.
- We validate various features of our system like vision, texting, alarm, location to ensure effectiveness in an actual situation.

2 LITERATURE REVIEW

The main goal of the wireless device is to provide women safety by sending alert text messages to the guardian's mobile device along with the user's location to raise an alarm for the user's rescue when she is under attack. Hence, the functionality and effectiveness of the device is crucial for providing user safety in those dangerous situations. There are many existing women safety system including wearable devices, smart lockets, smart footwear, smartphone applications and intelligent security systems. There is an intelligent women safety system [1], which consists of RFID tags and a microcontroller. The information is scanned with an RFID reader and passed on to the microcontroller after which the message is sent to the appropriate party. On the other hand, there is a smart foot device which is fitted to the user's shoes [2]. When activated, this device sends information about the user's location to the smartphone application.

However, this device can be turned on by unintentional foot tapping, thus very unreliable. Moreover, another IoT based product [3] named "Watch me", only uses a pulse rate sensor that immediately contacts the control room and closed one if excessive heart rate is detected. Unfortunately, it is unreliable as heart rate can rise or fall for any kind of physical activity. Just like Flex Sensor [4] that is a drop detection sensor. It is also unreliable as user can accidentally turn it on. Another device, that responds to three different kinds of commands: voice, click, and when it is forcefully thrown called Suraksha [5]. This device triggers the GSM module to send it's location to the predefined contacts when it is activated. Another existing solution is an alarm clock with a one-touch feature. Pushing a button triggers the device's GSM and GPS modules to send the user's location to the predefined SOS contacts [6].

Study [7] shows an embedded women safety system using Arduino and GSM module along with GPS to send an emergency text message with location and generate an alarm. This system sends a message to the police control room, family and friends, sends emergency call and also tracks the location of the user. This device comes with a shock generator, which can be used to deliver mild electric shocks to disable the attacker. This wireless device is very convenient and can be turned on just by the click of a push button.

Moreover, study [8] provides a multimodal cost-effective security paradigm for women who may be subjected to offensive risks using deep sensing technologies. Hidden Markov Models (HMMs) provide improved predictive analysis due to their dynamic probabilistic character, and they are useful for developing a dense sensing strategy by analyzing traces of suspicious activity. The use of face recognition and fuzzy labeling of vocal exchanges allows situation-based analysis of relative modeling. In the case of an aftershock, the GSM/GPS module will signal an emergency, otherwise, the female device carrier will receive a notification.

Another existing solution [9] is a wearable device prototype which analyzes physiological data and body posture. Here, body temperature and galvanic skin resistance are evaluated. An accelerometer with a triple-axis sensor is used to estimate body location. Once raw data is captured, a customized machine learning algorithm is used to recognize activities. Real-time data monitoring is possible by transmitting sensor data wirelessly to a cloud platform.

Study [10] shows a Smart Jewelry Bracelet, which is a wearable smart device mounted in a bracelet. It includes an Arduino-compatible Adafruit Flora controller with an accelerometer, gyroscope, pressure and temperature sensors, microphone, and a GPS. The sensors in the smart jewelry bracelet continuously collect data on the user's activity and vital signs. With Adafruit, a machine learning algorithm recognizes and distinguishes between the user's typical movements and any abrupt or sudden movements that might indicate an assault. When the bracelet detects an assault, it automatically contacts emergency contacts and pairs with the user's phone using Bluetooth. This smart jewelry bracelet was designed using field testing and user research.

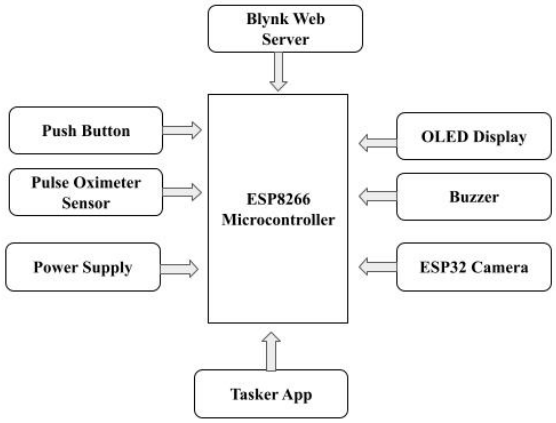

Figure 2: Block diagram of Wireless IoT device for women safety

3 PROPOSED METHODOLOGY

The main problem here is the unsafe condition that women face in the streets on a daily basis causing rape, assault, molestation, sexual harassment for which we feel that there is a need to provide an wireless IoT device for the users. By installing the ESP-32 application, the guardians can have access to the user's surroundings via live streaming and track her location via Tasker app.

Beside, the existing technologies in the present time do not apply a proper user-interface while the technologies that provide a user interface such as app and web interface are expensive and normal people cannot afford to buy one, therefore not economical. Considering the existing situation of products presently in the market and the growing need for women safety as crime rates continue to rise, we feel there is an urgent need to build an economically affordable women safety device that allows monitoring via remote access. This device is built keeping in mind of a women's defense abilities, effectiveness in a quick scenario and practical real-time application.

3.1 Objectives of the Proposal

Our project aims to develop a safety monitoring system to ensure the user's safety when she is outdoors. The system should be able to detect the unusual parameters (heartbeat and oxygen values). The guardian must be able to watch the user parameters like heartbeat and oxygen values in the Blynk IoT and be notified in their mobile device with alert text message with location in case of an emergency. We assume that the system should consist of the following elements such as ESP8266 microcontroller, buzzer, ESP-32 camera, pulse oximeter and heart rate sensor (MAX30101), Tasker app and the Blynk IoT, as illustrated in Figure 2.

Pulse oximeter and heart-rate sensor and the ESP- 32 camera provides information to the ESP8266 microcontroller. Through the Blynk web server, the Tasker App communicates with the mobile device of the guardian, who can monitor the user when she is outdoors. We adopt the following principles to our system:

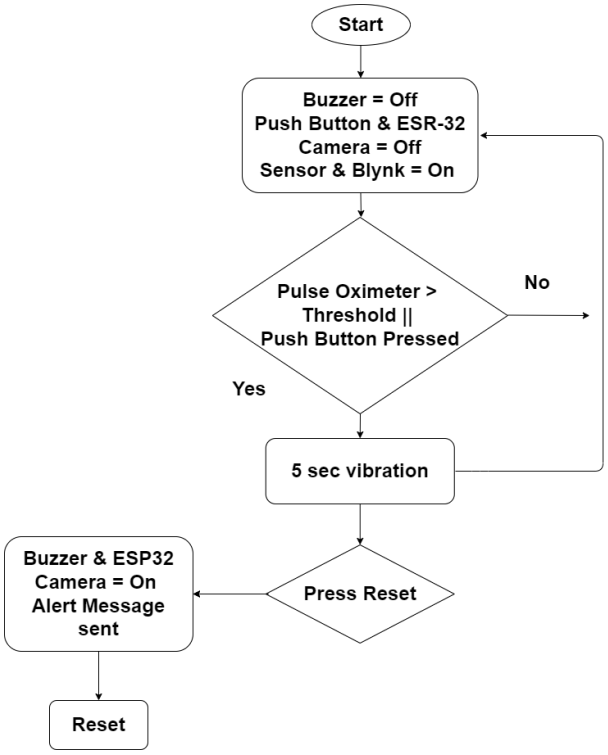

Figure 3: flowchart of our system

- It monitors user parameters (heartbeat and oxygen values).
- If there is any unusual reading, it activates the ESP-32 camera and notifies the guardian via alert text in their mobile device.
- Blynk web server displays the user parameters (pulse-rate and oxygen vales) on the interface, which can be accessed by the guardian using their mobile devices.
- The system monitors the user when she is outdoors.

3.2 Working Mechanism

Figure 3 is the flowchart showing the working mechanism of our women safety system. Based on the readings detected by the Pulse oximeter and heart rate sensor, the entire process occurs in two ways:

- Pulse Oximeter and Heart-rate Sensor = 1, If the Pulse Oximeter and Heart-rate Sensor is active, it means that an abnormal heart or oxygen saturation reading above the threshold value is detected. The ESP-32 camera wakes up from sleep mode and starts working. Then the ESP-8266 microcontroller will alert the buzzer. It will be high for 1 minute to alert the surroundings and an alert will be sent to the guardian through the application. The alert also can be seen in the web interface. The guardian has access to the video streaming record of the ESP-32 camera.
- Pulse Oximeter and Heart-rate Sensor = 0, If the Pulse Oximeter and Heart-rate Sensor detects normal readings of heartbeat and oxygen below the threshold value, indicating

the situation is normal. The values are sent to the Blynk web server and no alarm is raised. The system operates normally.

4 SYSTEM DESIGN

This section is divided into two parts. The first part is about user-driven design. The second part deals with system modeling.

4.1 User Driven Design (UDD)

In User-Driven Design, product development is guided by future users, designers and stakeholders. During the design process, every step is validated with the user. Generally, it is divided into two parts of product development: formulation of a problem and product development.

Problem Formulation: Our system would be used by young girls, women and guardians. When the system is implemented, guardians will be able to monitor the user and be alerted if any unusual activity is detected. The guardian can watch live video streaming via the ESP-32 web interface. The requirements of the system are user interface (e.g. application or web browser), monitoring the user's activity, and detection of unusual body parameters (heartbeat and oxygen level), (in Figure 4).

Product Development: In order to meet the requirements outlined in the problem description, the following tools are needed: ESP8266 microcontroller, buzzer, ESP32 camera, and pulse oximeter and heart-rate sensor. Next, flowchart, the block diagram, schematic diagram should be created for prototyping. Tests must be conducted after prototyping. Our tests show the system is useful to provide user safety by monitoring user activity that seems unusual. The system can alert the guardian through SMS notification along with the GPS coordinates that the user is in danger and also raising awareness to surrounding people by raising an alarm [19]. Hence, the user-driven design can be seen as a procedure of solving problems in multiple stages.

The features of our system are shown in Table 1. It covers both general and detailed features and some specific restrictions as well as some specific restrictions with the related technologies and algorithms.

- Detection of abnormal readings: The MAX30101 uses an internal LED to reflect light onto arteries and arterioles in the subcutaneous layer of the finger, which is then detected by a photodetector. This process is called photoplethysmography. This data is used by the MAX30101 to determine heart rate and blood oxygen saturation.
- Monitoring of user, Surveillance, and Face Recognition: The features listed here are part of the same section. The ESP32 camera is used to monitor the user (user's unusual activities). In addition to video streaming, it can also detect and recognize faces. If the Pulse oximeter and heart sensor detects any abnormal reading, then the ESP32 camera wakes from sleep mode and begins streaming video. The video can be viewed in the user interface.
- Communication: The system should be able to alert the user when an alarm condition occurs. The system must be able to communicate with the guardian regardless of the user's location. In case of an emergency, the system should effectively communicate with the guardian by sending alert notifications via text along with GPS location coordinates so that the user can be tracked. Here, the user interface acts as an intermediary between the user and the guardian, storing the user's vital parameters (heart-rate and oxygen values).
- Interface: The user interface connects the user to the system. The guardian can access the user data from anywhere and at any time. A web interface and application interface are created. The guardian can access user data and watch video streaming by either logging in to the application or by using a web browser.

5 SYSTEM IMPLEMENTATION

The section is comprised of two parts including a description of the hardware and a description of the software. The hardware section describes how the hardware is implemented and validated. The implementation section discusses the modules used in the system, whereas the validation section verifies its functionality.

The Figure 4 schema diagram illustrates the connections between the ESP8266 microcontroller and the other components such as ESP-32 camera module, heart sensor, oxygen sensor and the buzzer, where the symbols are denoted in Table 2.

5.1 Hardware Components

In our proposed architecture, we use four hardware components -

- ESP8266 microcontroller: The ESP8266 is a low-priced WiFi microprocessor integrated with TCP/IP networking. The ESP8266 microcontroller controls external sensors using its set of GPIO(general purpose input/output) pins. The ESP8266 can also be connected to a WIFI network, to the Internet, to a web server, and to a smartphone via WIFI.
- ESP-32 CAM: ESP32-CAM is a small, low-power camera module. It comprises a 2-megapixel OV2640 camera. ESP32-CAM is well suited for IoT applications including smart devices that upload images, facial recognition, video monitoring, etc. The ESP32 has built-in WiFi and Bluetooth modules.
- Pulse Oximeter Heart-rate Sensor: The MAX30100 Pulse Oximeter and Heart-rate Sensor measures the blood oxygen and heart-rate of the user. It works between the range of 1.8V and 3.3V power supply.
- Buzzer: The buzzer is an integrated electrical device that emits an alert sound to notify surrounding people. It has a frequency range of 30Hz - 65500 Hz and can vary from device to device.

5.2 Prototype

Figure 5 presents the prototype of our wireless women safety device. This operating model below illustrates the structure and functionality of the women safety system. The figure illustrates all the components involved in the system and how they engage with the user through the Blynk IoT web interface and Tasker app. The necessary conditions for installing the prototype are:

- An Internet connection is required.
- A power supply is needed.

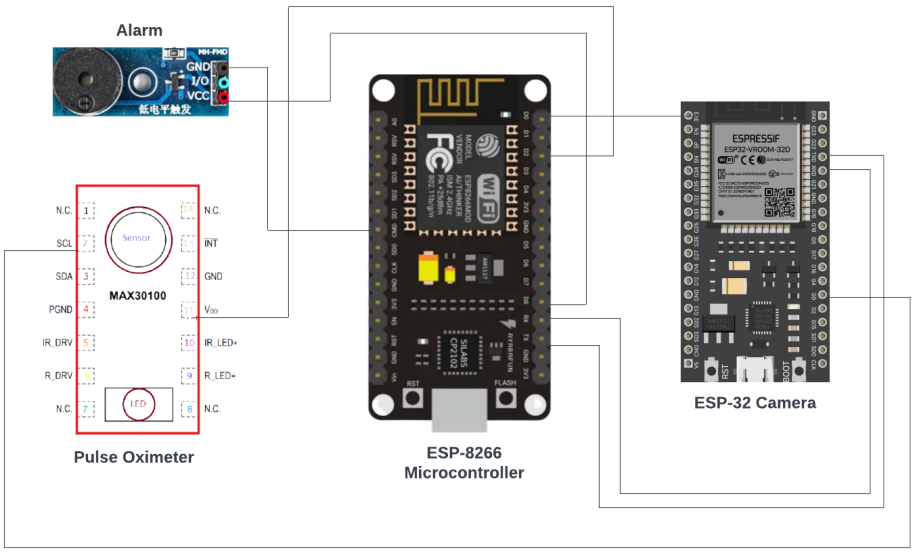

Figure 4: Schema Diagram of the proposed architecture

Table 1: Design and Implementation of Wireless IoT Device for Women's Safety

Functionalities		Particular Constraints	Used Technolgies	Existing Technologies
Detection of reading	<100Bpm >100Bpm	Validity >90% Validity >70%	Pulse Oximeter	PIR Sensor Temperature Sensor Skin Color Sensor
Monitoring of readings		High Validity	ESP-32 Camera	Web Camera SMOS
Surveillance		Coverage of user's surround- ings	ESP-32 Camera	Web Camera SMOS
Face Recognition		High Validity >80%	ESP-32 Camera	Nest Hello Mini OV 7690
Communication (Informing guardian)		High Validity >90%	Blynk web server, Tasker App	Blue - tooth GSM
Interface		High Validity >75%	Blynk web server	Webpage domain hosting

Table 2: The symbols indicator Table

Notation	Description
SG1	buzzer
GND	ground
Vin	input voltage
Tx	tranmitter pin
Rx	receiver pin

5.3 Web Interface

In our system, we used the Blynk web interface to connect the user (girl) and the guardian as shown in Figure 6. In interfacing, this prototype falls under the category of Human-to-Computer Interface (HCI). This interface allows the guardian to monitor her child using their mobile phone and track to see if they are in danger by monitoring the heartbeat and oxygen values and also opening the camera if necessary. Various types of interfaces are available, including Bluetooth, IP address connections, and APIs or Application Program Interfaces. In our working model, we used API and IP address connections. By entering the IP address in the web browser,

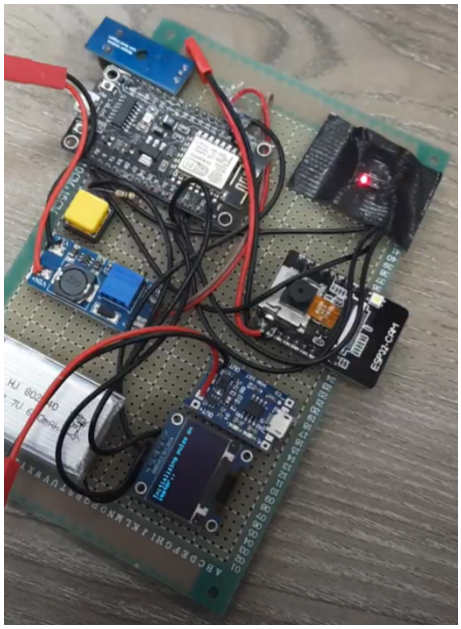

Figure 5: Prototype of wireless device

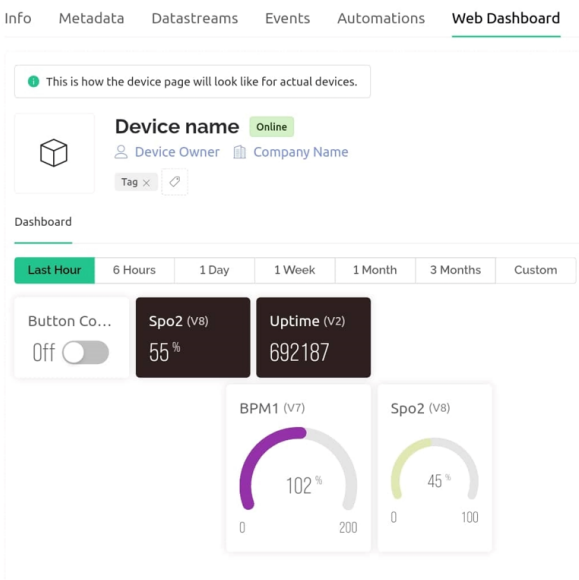

Figure 6: Blynk Web Interface

the user will be able to access the heartbeat and oxygen values via the Blynk web interface and watch video streaming using the ESP-32 camera.

The application that we implemented in our project is a Wireless Safety Device for Women. The users must download and register the Tasker app on their mobile device. This app acts as a communication medium between the user and the guardian. The Tasker app fetches the data from the Blynk web interface and sends the data to the guardian's mobile phone. In order for this to work, the guardian must log in to the system using the Wifi credentials (SSID and password). The guardian can access the user's parameters (heartbeat and oxygen values) by logging in to the Blynk web server.

In case of an emergency, the guardian is immediately notified with alert messages via text messages (through the Tasker app) showing the exact location of the user. The guardian can watch video streaming by simply opening our custom made website and clicking the start streaming option. The streaming can also be turned off manually.

6 SYSTEM VALIDATION AND RESULTS

We carried out several tests to checking whether every component in our system is working properly. Here is an analysis of the test results.

6.1 Test 1: Validation of Vision Functionalities

Aim: To examine whether the Pulse Oximeter Sensor is detecting the abnormal readings properly. Components: Pulse Oximeter and Heart-rate Sensor, ESP8266 microcontroller.
Components: ESP32 Camera and Blynk Web interface.
Objectives: The Pulse Oximeter and Heart-rate Sensor must be able to detect abnormal heart rate and pulse readings above the threshold value. If the user's heart rate is abnormally high, it should be high (HRT = 1), and if the user's heart rate is normal, it should be low (HRT = 0). Also, we need to check the delay time of the sensor. The delay time is the length of time that the heart sensor remains active after an abnormal heart rate reading is detected. The goals of using this ESP32 camera are as follows:

- Checking the resolution of video.
- Verifying the range of ESP32 camera.
- Checking the accuracy of the live video streaming in the user interface.
- Checking the accuracy of the live video streaming in the user interface

Procedure: The procedure is explained in the following steps:

- We established a connection between the ESP32 camera and the ESP8266 microcontroller.
- Once the connection between the ESP32 camera and the ESP8266 microcontroller is established, the network credentials for the connection between the system and the network (WiFi ssid and password) are typed.
- The user interface should display accurate live video streaming.
- While conducting the test, the attacker must remain at a distance of 1.5 meters from the camera.

Observations: The webcam image is shown in Figure 7. We can assess the quality of the video by looking at the picture (the video resolution is 240 HD). The camera is connected to the microcontroller and the web interface using the ESP8266. The user interface offers options for adjusting black and white, negative, and greyscale video streaming.
Results: HD video streaming is of average quality. In the web interface, the settings can be adjusted to enable black and white video

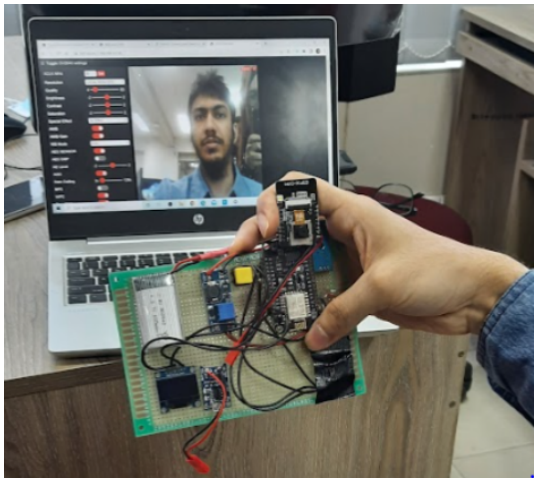

Figure 7: ESP-32 Camera results

streaming. The web camera covers a distance of three meters. Thus, we can conclude that the vision functionality is valid.

6.2 Test 2: Validation of Heart-rate and Oxygen Reading Detection Functionality

Aim: We examined whether the Pulse Oximeter and Heart-rate Sensor is detecting the abnormal readings properly (in Figure 8).
Components: Pulse Oximeter and Heart-rate Sensor,

Objectives: The Pulse Oximeter and Heart-rate Sensor will detect abnormal heart rate and pulse readings above the threshold value. If the user's heart rate is abnormally high, it will be high (HRT = 1), and if the user's heart rate is normal, it will be low (HRT = 0).
Procedure: Here is a detailed description of the procedure:

- Connect the Pulse Oximeter and Heart-rate Sensor to the ESP8266 microcontroller.
- Give power supply to the ESP8266 microcontroller and the Pulse Oximeter and Heart-rate Sensor will start working.
- There are two ways to detect readings:
  – Case 1: When the pulse readings are normal below 100Bpm.
  – Case 2: When the pulse readings exceed 100Bpm.

Observation: When the pulse readings went below 100Bpm and above 100Bpm the alarm automatically activated and the device started vibrating for 5 seconds.

6.3 Test 3: Evaluation of the buzzer notification system

Aim: We determined the frequency of the buzzer when it is active and to check the operation of the buzzer associated with the Pulse Oximeter and Heart-rate Sensor.
Components: ESP8266 microcontroller, Buzzer, Pulse Oximeter and Heart-rate Sensor.
Objectives: Pulse Oximeter and Heart-rate Sensor will activate the buzzer if abnormal readings far exceed the threshold value are detected.
Procedure: We conducted tests to determine whether the buzzer

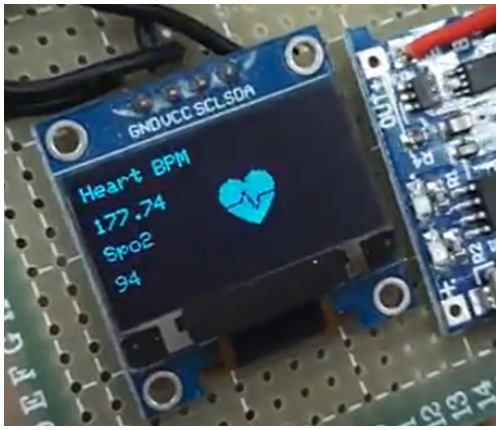

Figure 8: Display of BPM and Spo2 readings

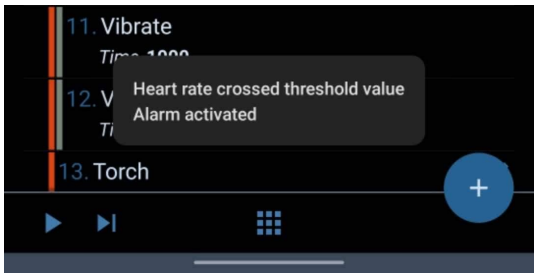

Figure 9: Alarm Activation after abnormal heart-rate.

is working properly. The following tests outline the procedure in detail.

- The negative pin of the buzzer is connected to the ground, while the positive pin is connected to pin 8.
- We connected the positive and negative of the Pulse Oximeter and Heart-rate Sensor to the ESP8266 microcontroller 5V and the ground.
- Case 1: If the pulse oximeter and heart-rate sensor detect abnormal readings above the threshold value, then we recorded the frequency of the buzzer and its state (high or low).
- Case 2: If the pulse oximeter and heart-rate sensor gave normal readings, then we recorded the frequency of the buzzer and its state (high or low).

Observations: The working mechanism of the buzzer is illustrated in Figure 9. The sounds could not be shown, however we can see the buzzer activation due to heart and pulse detection, as shown in Figure 9.
Results: Through this test, we can see if the buzzer is functioning in conjunction with the pulse oximeter and heart-rate sensor. It's on and off conditions. The buzzer is on when anomalous heart and pulse readings are detected and off when no abnormal heart and pulse readings are detected. In addition to scaring the attacker, the buzzer will also alert surrounding people. Hence, the operation of the buzzer is tested through this experiment.

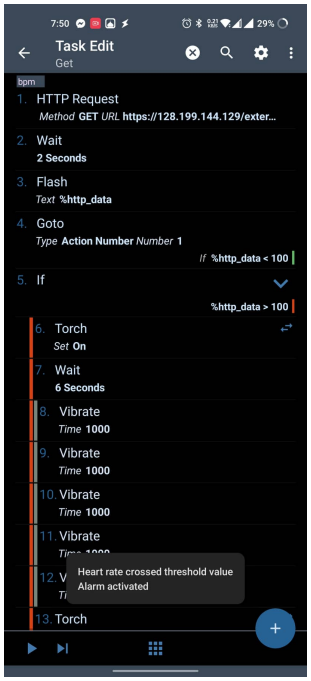

Figure 10: Alert condition in the Tasker application

6.4 Test 4: Authentication of Blynk Web Interface and Tasker Application

Aim: We verified the system's functionalities when all the parts are connected and checked the interface of the system.
Components: Buzzer, Pulse oximeter and Heart-rate sensor, ESP8266 microcontroller, ESP32 camera.
Objectives: The test aims to:

- The user interface (application, Blynk web interface) must allow users to access the data.
- In the event of an alert situation, parents/guardians must receive alert notifications (via Tasker app) that their child is in danger.

Procedure: Below is a brief explanation of the steps.

- We compiled the ESP8266 microcontroller code to provide the required credentials for connecting the ESP8266 microcontroller to the network.
- Following the system's connection to the network, we checked it's functionality in two ways:
  – Method 1 : The Tasker app displayed the output "Normal" when the pulse oximeter and heart-rate sensor show normal readings (HRT = 0). Using the Blynk web server, the user interface showed normal heart rate and oxygen readings of the user.
  – Method 2 : If the pulse oximeter and heart-rate sensor both indicate they are high, the ESP8266 microcontroller activates the buzzer. The guardian is notified about the alert by the Tasker application and the Blynk web browser shows the pulse and oxygen readings. Through the ESP-32 IP address, the guardian watches the video streaming. Here, the ESP-32 user interface streams live video.
- The guardian watches the live stream video by logging in the ESP-32 web interface.

Observations: If an alert condition exists (as shown in Figure 10), the guardian receives an alert message that the user is in danger along with her GPS location coordinates so that the user can be tracked. Starting and stopping video streaming options are available throughout the application interface and web interface, so the guardian can control video streaming.

6.5 Discussion

The tests are run on various components and each test is evaluated. Based on the results, some discussions are made. The first test results gives an idea of how the ESP32 camera works and the special features of the ESP32 camera. We also learn about video quality and how to implement black and white video streaming with an ESP32 camera.

The second test result sent by the pulse oximeter and heart-rate sensor shows its accuracy in detecting abnormal readings. The third test was done to inspect the response of a buzzer after detecting abnormal readings from the pulse oximeter and heart-rate sensor.

The fourth test is run through the Blynk web interace, which displays the heart-rate and oxygen values. The test also confirms the operation of the Tasker app which sends the alert text messages to the guardian's mobile device when abnormal heart-rate readings are detected. Based on the results of the test, these are the discussions that can be held. Table 4 below shows the components used in various tests and their results (OK/not OK). Besides, in Table 3, we present the comparison of our proposed architecture with existing technologies. Moreover, we present the accuracy of the gps receiver of our proposed approach in Table 5.

7 CONCLUSION AND FUTURE WORK

Furthermore, to conclude, this system reinforces the idea of gender equality by providing a self defense mechanism for women in the street in a dangerous situation as they are more prone to attacks. Our device is inspired by a real-life scenario faced by women. Considering the fact that a women can be attacked in multiple ways, we have tried to install a pulse oximeter and heart-rate sensor that measures heart-rate(pulse) and oxygen levels, both of which are very reliable parameters in case of an emergency. We have connected this device to a Blynk web server which displays these user parameters (heart- rate and oxygen values) to the server, from where the parent can access the data. This device also comes with an ESP-32 camera which can stream live video, and also has face detection to identify the attacker. We send text messages and location using the Tasker app. All these features make this device very well thought out and realistic to use, and ensures women's safety which is our main purpose in the research.

The device was integrated based on a real-life scenario faced by women in the streets. However, there are still some improvements to be made in our device. The future scope of our project is briefly described below:

Table 3: Comparison of Proposed Work with Existing Technologies

Name of Paper	Technology Used	Streaming	Stop Button	Reliable data
Hussain et al. (2016) [1]	RFID	No	No	Yes
Viswanath et al. (2016) [2]	Bluetooth	No	No	No
Helen et al. (2017) [3]	IoT	No	No	No
Priyanka et al. (2018) [4]	IoT	Yes	No	Yes
Bhardwaj et al. (2014) [5]	Machine Learning	No	No	Yes
Premkumar et al. (2015) [6]	IoT	No	No	Yes
Ruman et al. (2019) [7]	IoT	No	No	Yes
Seth et al. (2018) [8]	Hidden Markov Model	No	No	Yes
Jatti et al. (2016) [9]	Machine Learning	No	No	No
Patel et al. (2018) [10]	IoT	No	No	Yes
Our proposed architecture	IoT	Yes	Yes	Yes

Table 4: Table of test results

Components	Test 1	Test 2	Test 3	Test 4	Tested
ESP-8266 microcontroller	Yes	Yes	Yes	Yes	Yes
Buzzer	No	No	Yes	Yes	Yes
ESP-32 CAMERA	Yes	No	No	Yes	Yes
Pulse - Oximeter	No	Yes	Yes	Yes	Yes
Blynk Web and Tasker	No	No	No	Yes	Yes

Table 5: Accuracy of gps receiver

	Location shown in gps	Actual Location	Accuracy
Test Case 1 (Kallyanpur)	$23 \circ 47\prime 06.9N\ 90 \circ 21\prime 26.5E$	$23 \circ 47\prime 06.0N\ 90 \circ 21\prime 27.2E$	91%
Test Case 2 (Mirpur)	$23 \circ 48\prime 29.5N\ 90 \circ 21\prime 52.5E$	$23 \circ 48\prime 28.9N\ 90 \circ 21\prime 53.2E$	96%
Test Case 3 (Gulshan)	$23 \circ 79\prime 19.5N\ 90 \circ 21\prime 15.2E$	$23 \circ 79\prime 20.1N\ 90 \circ 21\prime 16.1E$	97%
Test Case 4 (Banani)	$23 \circ 80\prime 32.7N\ 90 \circ 41\prime 10.1E$	$23 \circ 80\prime 31.5N\ 90 \circ 41\prime 10.4E$	95%

- In our working model, we used a lithium-ion battery, which can be replaced by a polymer battery to reduce the size of the battery and the device.
- Other means of safety like PIR sensors or skin detection sensors can be added to the product, to detect alert situations.
- Installing a SOS Light indicator which will be flashing warning light thus continously drawing attention of the surrounding people that the user is in danger.
- The wearable device can also be made waterproof to prevent short circuiting.
- A voice sensor can be added which can be turned on specific keywords like "help". This feature can be built as a backup, because in a panic situation the automatic reflex of a women would make her scream or make any form of distress noise which is captured by the voice sensor.
- Buying a separate domain hosting and creating a user interface where only authorized people(guardians) can have access of user data, thus ensuring greater security.
- We may use machine learning in our device to improve the performance like automatically avoiding false alarms. The dataset can be analyzed online so that it can improve its performance based on users behavior.
- We will study the energy consumption related to our device in future.
- We will carry out experiments on real-users in the future to improve the usability of our device.

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